

Progress Report Update 3: NSU ClubHub

A Centralized Club Management & Engagement Portal

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Abstract—NSU ClubHub is a centralized platform designed to streamline club management and enhance student engagement at :contentReference[oaicite:0]index=0. Earlier development stages introduced a functional prototype and user interface improvements. Update 3 expands the system with role-based access control (RBAC), administrative dashboards, and analytics tools, transforming the platform into a scalable SaaS-style architecture. The system now supports Student, Club Admin, and Super Admin roles, enabling structured management of club activities, recruitment, and event coordination. These enhancements improve operational efficiency and lay the foundation for real data integration and secure authentication in future phases.

Index Terms—Club Management, Next.js, Role-Based Access Control, SaaS Dashboard, Student Portal

I. INTRODUCTION

University clubs foster leadership, collaboration, and professional growth, yet their management often suffers from fragmented communication and manual processes. NSU ClubHub addresses these challenges by providing a unified digital portal where students can explore clubs, track events, and apply for recruitment opportunities. Previous updates established the prototype and user interface improvements, while Update 3 introduces scalable architecture and administrative capabilities to support long-term system growth.

II. TOOLS AND TECHNOLOGY

The platform is built using Next.js 15 with React 19 and TypeScript, ensuring type safety and modern rendering capabilities. Tailwind CSS and shadcn/ui components provide responsive and consistent user interface design. Analytics visualization is powered by Recharts, enabling administrators to monitor application trends and engagement metrics. Currently, structured mock data is used for demonstration, with plans to integrate PostgreSQL databases and authentication mechanisms in future iterations.

III. KEY FEATURES AND IMPROVEMENTS

This update introduces role-based access control (RBAC), dividing system users into Students, Club Admins, and Super Admins. Students can browse clubs and submit applications through a structured multi-step form. Club Admins manage recruitment roles, applications, and event data through dashboards. Super Admins oversee platform operations and user management.

The SaaS-style dashboard redesign improves navigation and data visualization. Notification panels and responsive layouts enhance usability across devices. Analytics tools provide insights into club participation and application trends, supporting data-driven decision-making for administrators.

IV. PROGRESS AND IMPACT

Update 3 marks a transition from prototype functionality to a structured platform architecture. Administrative tools and analytics features improve operational efficiency and data monitoring. Role-based access ensures secure and organized system interactions. These improvements expand the platform's scope beyond basic club discovery to comprehensive management capabilities.

V. FUTURE PLANS

Upcoming development focuses on backend integration and security enhancements. PostgreSQL database implementation will enable real data storage and retrieval. NextAuth authentication will support secure login using institutional credentials. Automated email notifications and real-time updates will further enhance user engagement. These features aim to transform NSU ClubHub into a fully operational and scalable club management system.

VI. CONCLUSION

Progress Update 3 significantly advances NSU ClubHub by introducing scalable architecture, administrative dashboards, and analytics tools. The platform now supports structured club management and role-based operations. Future development will complete backend integration and real-time functionality, establishing NSU ClubHub as a comprehensive solution for university club engagement.